

BREAK WEEK · MIDPOINT RESET

Rest. Reflect. Begin.

You have completed five weeks of data science. This week you rest, look back at how far you have come, and begin planning the project that will define your Ngao Labs experience.

Duration: 1 Week No Assignments Capstone Launch

What You Have Built in Five Weeks

Week 1

Data Science Lifecycle

Bias, ethical issues, dataset exploration

Week 2

Python for Data Science

Pandas, NumPy, Matplotlib, data cleaning

Week 3

Preprocessing

Missing values, encoding, feature engineering

Week 4

Supervised Learning

Regression, classification, evaluation metrics

Week 5

Unsupervised Learning

K-Means, hierarchical clustering, silhouette score

Week 6

Neural Networks

Coming next

1. You Have Earned This Break

Five weeks of data science training while managing jobs, families, commutes, and the ordinary pressures of Kenyan life is not easy. Most learners started this programme not knowing what a DataFrame was. Most can now load a dataset, clean it, build a

supervised learning model, evaluate it properly, and explain what the confusion matrix is telling them. That is a real change, and it happened in five weeks.

Rest is the first purpose. The human brain consolidates learning during rest. The concepts from Weeks 1 through 5 will settle more firmly if you give yourself permission to stop engaging with the material for a few days. You will return to Week 6 faster and clearer than if you spend the break grinding.

Reflection is the second purpose. Before moving into the second half of the programme, look back at what you built. Open your Week 4 notebook. Look at your confusion matrix. Read a comment you wrote in your code. You are not the same learner who started Week 1. Noticing that matters.

Capstone ideation is the third purpose. Not execution, ideation. You are not expected to build anything this week. You are expected to begin thinking seriously about what problem you want to solve, who it affects, and why it matters.

What This Break Week Is Not For

- Not a catch-up sprint. If you missed assignments in Weeks 1 through 5, use one or two days to address the most critical gaps. Do not attempt to compress three weeks of content into seven days of break. That is the fastest route to burning out before the second half begins.
- Not a Week 6 preview. Do not start the neural networks content early. The break is a boundary, not a runway.
- Not a reason to disengage. The programme continues in one week. Your mentor is already waiting to hear from you.

Tip: The single best thing you can do this break week is read the Week 10 lesson on Moodle. It contains the complete capstone guide. Reading it now, before you start building, means you will make better decisions throughout Weeks 6 through 9. The learners who read it early almost always produce stronger final projects.

2. What You Have Built — A Recap of Weeks 1 to 5

Week 1: You mapped the data science lifecycle from problem definition to deployment, explored how AI is being used in Kenyan healthcare, agriculture, and finance, and looked directly at how bias entered a facial recognition system trained without sufficient data from darker-skinned faces. You began with the Financial Inclusion in Africa dataset, a dataset you will encounter again at Week 9.

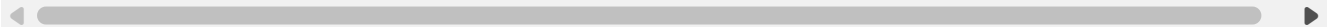
Week 2: You learned Python for data science. You can now load a CSV file into Pandas, summarise it, clean it, write functions that apply transformations to columns, and produce line charts and histograms in Matplotlib. Every technique built in Weeks 3 through 10 depends on these foundations.

Week 3: You learned how to prepare data for modelling. Missing value imputation, outlier handling, one-hot encoding, normalisation, and feature engineering are the steps that happen between raw data and a trained model. You practised on real datasets from Tanzania and Nigeria.

Week 4: You built your first supervised learning models. A regression model to predict a continuous value, and classification models to predict a category. You evaluated them properly: not just accuracy, but precision, recall, F1, and AUC-ROC. You saw the difference between a model that overfits and one that generalises.

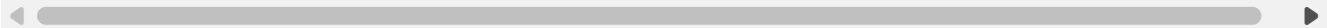
Week 5: You stepped into unsupervised learning. Without any labels, K-Means clustering found natural groups in data using only distance. You chose the right number of clusters, visualised them, and explained what each cluster represented in plain language, a skill that matters at the presentation stage of the capstone.

Area	Skills Available Now
Data Handling	Pandas, NumPy, CSV loading, df.describe(), df.info(), df.isnull()
Visualisation	Matplotlib histograms, scatter plots, line charts, bar charts
Preprocessing	Imputation, encoding, scaling, feature engineering
Supervised Learning	Linear regression, logistic regression, decision trees, k-NN
Evaluation	Accuracy, F1, AUC-ROC, RMSE, MAE, confusion matrix, cross-validation
Unsupervised	K-Means, hierarchical clustering, silhouette score



What Is Still Coming

Coming Week	What It Adds to the Capstone Toolkit
Week 6	Neural networks, deep pattern recognition on structured data
Week 7	Generative AI, text generation, summarisation, Swahili NLP
Week 8	Random Forest, XGBoost, time series, recommendation systems
Week 9	Responsible AI, bias audit, fairness criteria, KDPA 2019



Note: Your capstone idea does not need to wait for Week 8 techniques. A well-scoped problem that uses logistic regression or K-Means thoughtfully is a stronger project than an ambitious plan to apply neural networks to data you have not yet explored. Start with what you know. Upgrade the model as the weeks progress.

3. The Capstone Project — What It Is and Why It Matters

The capstone is not a test of whether you can build a perfect model. It is a demonstration that you can take a real problem, gather data, apply the appropriate technique from your toolkit, evaluate the result honestly, document the process professionally, and explain it to an audience that includes people who are not data scientists.

The capstone runs from Week 6 to Week 10. It is the largest single deliverable in the programme and is worth 40% of the total grade. It is also the only deliverable that becomes a permanent public artefact: a GitHub repository that future employers, collaborators, and Ngao Labs alumni can find, read, and reference.

The Five Capstone Tracks

Learners choose one track. Each track matches a broad problem type to the techniques taught across the programme:

- **Classification:** Predict which category something belongs to. Loan default, disease detection, sentiment, news topic. Uses techniques from Weeks 4 and 8. The most common track and the one with the most available Kenyan datasets.
- **Regression and Time Series:** Predict a numerical value, either for a single observation or as a sequence over time. Crop yield prediction, house price estimation, M-Pesa agent float demand forecasting, weekly rainfall prediction. Uses techniques from Weeks 4 and 8.
- **Clustering:** Find hidden groups in data without labels. Customer segmentation, constituency socioeconomic profiling, topic discovery in text. Uses techniques from Week 5. Produces a different type of output than the other tracks: insights about group structure rather than predictions.
- **Recommendation:** Match users to items they have not seen. Job recommendations, product suggestions, content discovery. Uses techniques from Week 8. Requires data with user-item interaction history.
- **Generative AI:** Use a large language model to generate, summarise, or transform text. Swahili agricultural advisory generation, report summarisation, document translation. Uses techniques from Week 7. The most distinct track: evaluation is qualitative as well as quantitative.

The Capstone Rubric — Brief Summary

The full rubric is in the Week 10 lesson. The five criteria are:

1. **Problem Definition and Scope (8 marks):** Specific problem statement with a clear target variable, population, data source, and metric. Supported by a datasheet for the dataset.
2. **Data Science Workflow (12 marks):** Four clean, runnable notebooks covering EDA, preprocessing, modelling, and evaluation. A baseline comparison is required.
3. **Evaluation Quality (10 marks):** Two metrics, disaggregated by at least one demographic or geographic variable, plus an error analysis.
4. **Documentation and GitHub (8 marks):** A professional README, clean notebooks, and a complete requirements file.
5. **Responsible AI Statement and Presentation (8 marks):** All five impact assessment questions answered specifically, a named fairness criterion, and a 5 to 8 minute presentation.

Total: 46 marks, scaled to 40% of the programme grade.

The Week 10 lesson contains the complete project guide: how to write a strong problem statement, how to structure your GitHub repository, how to run an error analysis, how to write the Responsible AI Statement, and how to present data science work to a mixed audience. Read it this week before you submit your capstone idea to your mentor. It will make your idea sharper.

Read the Full Capstone Guide

The Week 10 lesson contains the complete capstone guide: dataset selection, project workflow, GitHub setup, evaluation requirements, the Responsible AI Statement template, and the assessment rubric. Read it this week, not in Week 10.

[READ THE FULL CAPSTONE GUIDE](#)

4. Your Capstone Mentor

Starting this week, every learner is assigned a Capstone Mentor. Your mentor is a data science practitioner, an industry professional, a researcher, or a Ngao Labs alumnus who has volunteered to guide one or more learners through the capstone journey. Your mentor is not your grader. Your mentor is your thinking partner.

Who Your Mentor Is

Mentors come from across Kenya's data science and technology ecosystem: analysts working at Kenyan banks and insurance companies, researchers at organisations like APHRC and CGIAR, product managers at Nairobi tech startups, and Ngao Labs alumni who completed the programme in a prior cohort. Every mentor has agreed to Ngao Labs' mentorship standards: responsiveness, honest feedback, and a commitment to helping you succeed.

How to Find Your Mentor

Your mentor assignment is published in the Moodle "Capstone Mentor Assignments" page. Your mentor's name, organisation, and preferred contact method are listed there. Check this page before the end of the break week. If your assignment is missing or there is an error, contact the Ngao Labs team immediately at the support email in your Moodle dashboard.

What the Mentor Relationship Looks Like

The mentor relationship has a defined structure. It is not informal or open-ended. Both you and your mentor know what is expected at each stage:

- **Break Week:** Your mentor reads your background and waits for your first contact. You send an introduction message and, before the end of Week 6, your project idea document for review.
- **Week 6:** Your mentor reviews your project idea and responds within three working days with approval, a revision request, or a suggestion to pivot to a different direction.
- **Week 7:** Your first formal check-in session, video call or voice note exchange on the agreed platform. You share your EDA findings and your first model results. Your mentor gives feedback on your approach.
- **Week 8:** Your second formal check-in. You share your improved model, your preliminary evaluation, and any questions about your ensemble or time series approach.
- **Week 9:** Your third formal check-in. You share your Responsible AI Statement draft and discuss the fairness findings from your disaggregated evaluation. Your mentor confirms you are on track for Week 10.
- **Week 10:** Your mentor attends or watches your final presentation. They may ask one or two questions during the Q&A session.

What to Bring to Every Mentor Session

Do not enter a mentor session with nothing prepared. Your mentor is giving their time voluntarily. Prepare:

- One specific question about your project that you cannot answer alone.
- A brief status update: where you are in the workflow, what you completed since the last session, what is blocking you.
- Your current notebook or results, shared before the session if possible.

- One specific thing you want feedback on: a feature engineering choice, a model comparison, a section of the Responsible AI Statement.

The Mentor's Role Versus Your Role

What Your Mentor Does	What You Do
Reviews your project idea and approves it	Writes the idea document and submits it
Gives feedback on your approach	Makes the technical decisions
Flags risks you may not have seen	Investigates and responds to flagged risks
Connects you to resources and datasets	Does the EDA, modelling, and evaluation
Confirms you are on track for Week 10	Builds the project, week by week
Asks questions at the final presentation	Answers questions from a position of knowledge

Warning: Your mentor is not your co-developer. Do not send your notebook to a mentor and ask them to explain why it is not working. Debug first. Try at least two things to fix the problem yourself. Then send your mentor a message that says: "I encountered this specific error. I tried X and Y. Neither worked. Here is what I think may be happening. What would you check next?" That is how professionals ask for help.

Note: If you do not hear from your mentor within three working days of sending your idea document, follow up once. If there is still no response after a second follow-up, contact the Ngao Labs team. Mentors are volunteers; sometimes the first message gets missed. Escalate calmly rather than assuming you have been forgotten.

5. Choosing Your Capstone Direction

The single most important decision of your capstone is the problem you choose to work on. A well-chosen problem makes every subsequent decision easier: the dataset is obvious, the target variable is clear, the metric follows naturally, and the Responsible AI Statement writes itself because you understand who the model affects.

A poorly chosen problem creates friction at every stage. You spend Week 6 not doing EDA but searching for data. You spend Week 8 not improving your model but questioning whether the target variable makes sense.

What a Good Capstone Problem Looks Like

It is specific. "Financial inclusion in Kenya" is a theme. "Predicting whether a Kenyan adult owns a bank account based on household demographic characteristics, mobile phone access, and primary occupation type, using the Financial Inclusion in Africa dataset, measured by AUC-ROC" is a problem. Specificity is the difference between a project that impresses and one that drifts.

It is Kenya-grounded. The problem should connect to a real challenge, institution, population, or dataset in Kenya or East Africa. If your model's results are relevant to an actual organisation like HELB, Equity Bank, KALRO, the Kenya Red Cross, a Nairobi hospital, a county government, say so explicitly. Domain relevance makes the Responsible AI Statement richer and the presentation more compelling.

It is achievable in five weeks. You have 25 weeks of free time spread across five working weeks, roughly. You are also learning neural networks, generative AI, and ensemble methods in parallel. Scope your project to what one person can do in that time: one target variable, one dataset, one primary metric, one model family, and one meaningful evaluation.

It is ethically grounded from the start. Before you choose a problem, ask: who does a model trained to solve this problem make decisions about? What is the worst plausible outcome if the model makes systematic errors? Who bears the cost of those errors? If you cannot answer these questions in two sentences at the ideation stage, the problem is not specific enough.

Things to Avoid When Choosing a Problem

- Avoid "prediction for prediction's sake". If you cannot name who would use your model's output and what decision they would make differently because of it, the problem has no stakeholder and the capstone will have no real-world grounding.
- Avoid data-first thinking. Do not choose a problem because a dataset exists. Choose the problem first, then find the dataset. If the best dataset for your problem is unavailable, revisit the problem, but start with the problem.
- Avoid overly academic problems. Reproducing a published machine learning paper is an academic exercise, not a portfolio project. Employers want to see that you identified a real problem, gathered appropriate data, and made defensible technical decisions. That is different from implementing someone else's method.
- Avoid problems that require data you cannot ethically obtain. Building a facial recognition system using scraped social media images, scraping personal data from Kenyan government portals, or processing medical data without appropriate consent are all off-limits regardless of the technical interest of the problem.

Kenyan Problems That Make Strong Capstone Projects

These are starting points, not assignments. Use them as inspiration:

- **Agriculture and food security:** Predicting crop yield by county using satellite-derived NDVI data and KNBS agricultural survey data. Classifying smallholder farmers by risk profile using mobile money and weather data. Forecasting monthly maize prices in Nairobi using KNBS commodity price records.
- **Financial services:** Predicting loan default using demographic and transaction history features, Zindi Loan Default dataset. Classifying M-Pesa agents by float demand risk using transaction pattern features. Profiling unbanked Kenyans using the Financial Inclusion in Africa dataset clustering approach.
- **Health:** Predicting NHIF claim amounts by facility type and patient demographics. Classifying disease risk by geographic cluster using KEMRI survey data, where available and licensed. Segmenting county health facility performance using K-Means on MoH service volume data.
- **Education:** Classifying KCPE score bands by school and constituency using KNBS and KNEC data. Predicting student retention using HELB loan repayment pattern features. Clustering Kenyan secondary schools by socioeconomic context to identify resource equity gaps.
- **Transport and urban:** Predicting Nairobi traffic incident frequency by road segment using NTSA open data. Forecasting matatu passenger demand by route using mobile phone data proxies. Clustering Nairobi constituencies by road infrastructure quality using satellite data.
- **Language and media:** Classifying Swahili news articles by topic, the Week 7 Zindi challenge extended to a custom dataset. Summarising English-language Kenyan policy documents into Swahili using a large language model, Generative AI track.

Tip: The strongest capstone ideas in past cohorts were not the most technically ambitious. They were the ones where the learner could say in one sentence why the problem matters to a specific Kenyan community, and could name a real organisation that would care about the results. Domain clarity produces better models, better documentation, and better presentations.

6. Writing Your Project Idea Document

The project idea document is not a formal research proposal. It is a single-page document, around 300 to 500 words, that tells your mentor enough to evaluate whether your project is achievable, well-scoped, and appropriate for a 5-week capstone. It is the first thing your mentor reads. Write it clearly.

The Four Required Elements

Your document must answer four questions explicitly:

1. **What is being predicted or discovered?** Name the target variable precisely. For classification: the category you are predicting and how many classes it has. For

regression: the numerical value and its unit. For clustering: what kind of groups you are looking for and how you will interpret them. For generative AI: what type of text will be generated and in what language.

2. **For which population?** Name who the model makes decisions about. Not "Kenyan people" but "smallholder maize farmers in Rift Valley counties" or "NHIF claimants in Level 4 and Level 5 public hospitals". The more specific the population, the easier it is to write a Responsible AI Statement.
3. **Using what data?** Name the dataset: its source URL, number of rows and columns, and which column is the target variable. If you have not yet downloaded it, explain where it is available and why you believe it is accessible. If you plan to combine two datasets, name both and explain the join key.
4. **Measured how?** Name your primary evaluation metric and explain in one sentence why it is the right metric for this problem. If you are unsure, say so and explain your reasoning. Your mentor will help you finalise the metric choice.

Additional Elements That Strengthen an Idea Document

- One sentence on the real-world context: who would use this model's output? An analyst at Equity Bank? A KALRO agronomist? A county health director? Naming a plausible user grounds the project in practice.
- One sentence on the capstone track: which of the five tracks does this belong to? This helps your mentor assess whether the technique is appropriate for your problem type.
- One sentence on a known risk: what is the biggest uncertainty in your plan? The dataset might be too small. The target variable might be noisy. You might not have access to a key column. Name it. Mentors respect honesty about uncertainty more than false confidence.

Idea Quality Checklist

Target variable named precisely, not just "something about loans"

Population named specifically, not "Kenyans" or "African people"

Dataset identified with source URL and approximate size

Primary evaluation metric named with one-sentence justification

Real-world context: who would use this model's output?

One known risk acknowledged honestly

The Problem Statement Test

After writing your idea document, extract the core of it into one or two sentences that answer all four required elements. If you cannot write that sentence, the idea is not specific enough yet. Do not submit the idea document until you can pass this test.

Example of a passing problem statement: "Using the Financial Inclusion in Africa dataset (Zindi, 23,524 rows), predict whether a Kenyan adult owns a bank account based on household demographic characteristics, mobile phone access, and primary occupation type, measured by AUC-ROC, disaggregated by urban/rural location type."

Example of a failing problem statement: "Using machine learning and AI to improve financial inclusion outcomes for underbanked populations in Sub-Saharan Africa."

The first has a target variable, a population, a dataset, and a metric. The second has none of these. If your idea document produces the second type of statement, revise before submitting.

Warning: Do not submit your first draft. Write the idea document, sleep on it, read it again the next day, and then ask yourself: if I were a mentor seeing this for the first time, would I know exactly what this learner is trying to build? If the answer is no, revise.

7. The Approval Process

Submit your project idea document to your mentor by the end of Week 6. Instructions for submission are in the Moodle "Capstone Idea Submission" activity on your dashboard. You will submit the document there, and your mentor will receive a notification.

What Your Mentor Reviews

Your mentor reads your idea document with four questions in mind:

1. Is the problem specific and testable? Can a model trained on the named dataset actually predict the named target variable? Does the problem statement pass the four-element test?
2. Is the scope realistic? Can one learner, in five weeks of part-time work while also studying Weeks 6 through 9 content, complete this project to the standard required by the rubric?
3. Is the dataset accessible? Is the named dataset downloadable today, at no cost, without requiring an institutional data sharing agreement or a Kenya government data request?
4. Are there ethical red flags? Does the project involve personal data that requires KDPA compliance? Does the proposed model make decisions about people in a way that requires specific fairness analysis? These are not disqualifying, but your mentor needs to flag them early.

The Three Outcomes

- **Approved:** Your mentor agrees the idea is well-scoped and you can proceed. Begin your EDA immediately. Do not wait for additional confirmation before downloading the data.
- **Approved with conditions:** Your mentor approves the general direction but requires one specific change before you proceed: a narrower population, a different dataset, a more appropriate metric. Address the condition and confirm with your mentor before starting the EDA.
- **Revision requested:** Your mentor believes the idea needs significant rethinking. This is not a rejection. It is a redirect. Your mentor will explain specifically what is wrong and what direction would work better. Revise and resubmit within two days.

How Long Approval Takes

Your mentor will respond within three working days of receiving your submission. If you submit your idea document on Day 1 of Week 6, you should have an approval decision before the end of Day 4. This gives you the remaining days of Week 6 to begin your EDA.

What to Do While Waiting for Approval

Do not sit idle. While you wait for your mentor's response:

- Download the dataset you proposed and open it in Python. Run `df.head()`, `df.info()`, and `df.isnull().sum()`. You do not need approval to explore data, just to proceed with formal modelling.
- Read the Week 10 lesson if you have not already done so. The problem statement template, notebook structure, README template, and Responsible AI Statement guide are all there.
- Open a GitHub account if you do not have one. Create a public repository named after your project. Push an empty README. The habit of committing work to GitHub must start in Week 6, not Week 9.

What to Do If Your Idea Is Rejected

It is rare for a mentor to reject an idea outright. When it happens, it is almost always because the problem is either too vague to execute or requires data that does not exist or cannot be used ethically. In either case, your mentor will tell you what kind of problem would work instead.

Your contingency plan, required in the idea document, is for this situation. If your primary idea is rejected, your mentor and you agree on a pivot using your contingency direction. Do not panic. A pivot in Week 6 is far better than a collapse in Week 8.

Tip: The approval process is a quality gate, not an obstacle. Every idea that clears the gate is one that your mentor believes you can complete to a high standard in

five weeks. Trust the process. If your mentor asks you to change something, they are protecting you from a harder conversation in Week 9.

8. The Incremental Development Plan

The capstone is not built in one week. It is built incrementally, one layer at a time, across five weeks. Each week adds a new layer to what you built the week before. The goal at the end of every week is a working notebook, not a finished project, but a documented step forward.



The Week-by-Week Development Rhythm

- **Break Week:** Ideation and mentor introduction. Output: a drafted project idea document ready to submit in Week 6. Contact your mentor. Introduce yourself. Tell them your idea is coming.
- **Week 6:** Idea approval and EDA. Submit your idea document by Day 1 or 2 of Week 6, do not wait until Day 5. Once approved, download the dataset and begin your EDA notebook. By the end of Week 6, you should have: a GitHub repository, a README with your problem statement, and a completed EDA notebook that answers the seven required questions from the Week 10 lesson.
- **Week 7:** Preprocessing and first model. Complete your preprocessing pipeline and train your first model on the cleaned data. This does not need to be your best model; it needs to be a working model with a baseline comparison. Post your project plan to the Capstone Peer Review Forum by the end of Week 7.
- **Week 8:** Model refinement and improvement. Apply the Week 8 techniques, ensemble methods, time series, or recommendation systems, to improve your first model. Run your disaggregated evaluation. Complete your second check-in session with your mentor and show them your current results.
- **Week 9:** Responsible AI Statement and ethical review. Draft your Responsible AI Statement using the framework from the Week 9 lesson. Run your error analysis. Identify and name the subgroup where your model performs worst. Complete your third check-in with your mentor. Show them your Responsible AI Statement draft.
- **Week 10:** Final polish and presentation. Clean your notebooks, write your final README, complete the Responsible AI Statement, and prepare your presentation. Submit everything to the Moodle Capstone Assignment box by the end of Week 10.

The Check-In Rhythm

Three formal check-in sessions with your mentor across Weeks 7, 8, and 9. The format is flexible: video call, voice note exchange, or written update by message, but the timing is fixed. Missing a check-in does not exempt you from the feedback your mentor would have given. It means you miss that feedback entirely.

Show what a well-prepared check-in message looks like:

```
Week 7 Check-In Message Example: Hello [Mentor name], Here is my Week 7 update. My project is predicting whether Kenyan adults own a bank account using the Financial Inclusion dataset. What I completed this week: EDA (attached), preprocessing pipeline, and a first logistic regression classifier. Current performance: AUC-ROC of 0.76 vs baseline of 0.62. What I am working on next: trying a Random Forest and comparing. My question for you: the mobile money usage feature has 18% missing values. I imputed with the median. Is there a better approach for a feature this important to the problem?
```

This message is 100 words. It covers what was done, what the numbers are, what comes next, and one specific question. It respects the mentor's time.

Warning: Do not disappear between check-ins and then send a long message in Week 9 asking for help with everything at once. Your mentor can help you most when problems are caught early. A question about feature engineering in Week 7 is solvable. The same question in Week 10, with a deadline in 48 hours, is a crisis.

The GitHub Commit Habit

Your repository's commit history is part of your submission. A repository that was last committed at Week 6 and then shows nothing until Week 10 fails the "evidence of iterative work" requirement of the assessment rubric. Commit something every time you make meaningful progress: a completed EDA cell, a new feature column, a trained model, an evaluation table.

You are not expected to commit polished work every day. You are expected to commit real work at regular intervals. Reviewers and employers look at commit history. An active, well-labelled commit history tells a story of a practitioner who works consistently. That is what you are building.

9. What Makes a Great Capstone Project

The difference between a good capstone and a great one is not the complexity of the model. It is never the complexity of the model. The capstone projects that stand out share three qualities that have nothing to do with which algorithm was used.

Specificity Throughout

From the problem statement to the Responsible AI Statement to the final presentation, the best capstone projects are specific at every level. The problem statement names a population. The EDA produces a plot that reveals something specific and unexpected about that population. The error analysis names the specific subgroup where the model fails and the specific reason it fails. The Responsible AI Statement names the specific harm that could result from a false positive in this domain.

Specificity is harder than generality. It requires that you understand your problem deeply enough to make claims that are true, testable, and connected to the real world. That understanding comes from reading about the domain, not just running the code.

Honest Evaluation

The projects that most impress professional reviewers are not the ones with the highest AUC-ROC. They are the ones where the learner reports their results honestly: "My overall AUC-ROC is 0.81, but for rural applicants it is 0.67. I have not been able to close that gap with the data available." That sentence tells a reviewer more about the learner's ability than any impressive metric ever could.

Honest evaluation requires running the disaggregated evaluation before you write the Responsible AI Statement, not after. It requires looking at the error analysis and reporting what you find, even when what you find is uncomfortable. It requires saying in the presentation: "Here is what this model cannot do."

Documentation That Respects the Reader

The README is the first thing a reviewer reads. If the README is a single paragraph that says "This is my machine learning project", the reviewer closes the tab. If the README answers the six required questions from the Week 10 lesson: what the problem is, who it affects, what data was used, what methods were applied, what the results were, and how to run the code, the reviewer reads the notebooks.

Clean, commented notebooks tell the same story. A notebook where every cell has a Markdown header explaining what it does and why, where every plot has a title and axis labels, and where every result is followed by one sentence interpreting it in plain language is not harder to build. It is the same technical work with a layer of communication on top. That layer is what separates a working notebook from a portfolio piece.

Tip: Set a rule for yourself from Week 6: every time you write a cell of code, write one Markdown cell above it that explains what the code is doing and why you

made that specific technical choice. Not what the code does, why you made the decision. Those cells become the skeleton of your presentation script.

Three Habits to Start This Week

1. Open a GitHub repository today. Name it after your project. Push an empty README with your problem statement draft as the first line. The habit of committing to GitHub must start in the break week.
2. Read the Week 10 lesson in full. Not for modelling advice, for the README template, the Responsible AI Statement template, and the submission checklist. Build your project toward those requirements from Day 1, not backward from them in Week 10.
3. Send your mentor an introduction message today or tomorrow. It does not need to be long. Introduce yourself, mention your programme background, and say that your project idea document will be with them by the start of Week 6. That one message tells your mentor you are organised and professional.

10. Before Week 6 Begins — Your Break Week Checklist

Close the lesson with a concrete, actionable summary of what each learner should do during the break week. Frame this as a sequence, not a parallel list: some items depend on others.

1

Rest (Day 1 to 3)

Take at least two full days away from the learning material. No notebooks, no Zindi, no Discord study groups. Rest is not a reward for completing five weeks; it is a strategy for performing well in Weeks 6 through 10.

2

Read the Week 10 Lesson (Day 2 or 3)

Open the Week 10 lesson on Moodle. Read it completely. Make notes on: which track you are drawn to, what the submission checklist requires, and what the Responsible AI Statement must cover. This will take around 45 minutes and will inform every subsequent decision about your capstone.

3

Find Your Mentor (Day 3 or 4)

Open the "Capstone Mentor Assignments" page on Moodle. Find your mentor's name, organisation, and contact details. Send them a short introduction message within 24 hours of finding their details. Do not wait until the end of the break week.

4

Draft Your Project Idea (Day 4 to 6)

Choose a direction, identify a dataset, and write the first draft of your project idea document. Use the four required elements and the idea quality checklist from

Section 6. Sleep on the first draft before revising it.

5

Create Your GitHub Repository (Day 5 or 6)

Create a public GitHub repository named after your project. Push a README with your problem statement draft as the title and first paragraph. Share the repository link in your introduction message to your mentor.

6

Revise and Finalise the Idea Document (Day 6 or 7)

Read your idea document again. Run it through the idea quality checklist. Pass the problem statement test from Section 6. If it passes, you are ready to submit. If it does not, revise until it does. Submit the idea document at the start of Week 6, not on the last day.

Note: You do not need a perfect idea. You need a specific, scoped, and honest idea that your mentor can evaluate fairly. Perfect ideas do not exist at the ideation stage. Specific ideas do.

BREAK WEEK ACTIVITIES

Required (Complete Before End of Week 6)

1. Send an introduction message to your assigned mentor.
2. Submit your project idea document via the Moodle "Capstone Idea Submission" activity. Deadline: end of Week 6, Day 2.
3. Create your public GitHub repository and add your problem statement as the first line of the README. Share the link with your mentor in your introduction message.

Optional (If You Have Time and Energy — Do Not Do These Instead of Resting)

- **Catch-up:** If you have incomplete lab submissions from Weeks 1 through 5, address the most impactful gaps. Facilitators are available on the Moodle discussion board during the break week. Prioritise Week 3 and Week 4 content: preprocessing and evaluation are the most critical foundations for the capstone.
- **Dataset exploration:** Browse the Zindi challenge archive and the KNBS open data portal for datasets relevant to your idea. Download one candidate dataset and run `df.info()` and `df.describe()` on it. Do not start modelling.
- **Week 10 preview:** Read the Week 10 lesson if you have not already done so. Focus on the problem statement template, the GitHub README template, and the Responsible AI Statement structure.

- **Peer connection:** Post a brief introduction in the Break Week forum, see below. Tell your cohort what direction you are considering for the capstone. Reading your peers' ideas often clarifies your own.

What Not to Do During the Break

- Do not start Week 6 content early.
- Do not attempt to retrain a model from a previous week for fun.
- Do not spend more than 3 hours per day on any programme-related activity.
- Do not let capstone ideation consume the rest days.

BREAK WEEK FORUM

Break Week: Rest Check-In and Capstone Directions

This is a low-stakes, optional forum. There is no required response or reply deadline. Post if you want to connect with your cohort. Do not post if you are using the time to rest.

Three suggested post types:

1. A rest check-in: one sentence on what you are doing this week that has nothing to do with data science. This is a community post, not a progress report.
2. A capstone direction draft: share your problem statement draft and ask your peers if it sounds specific and scoped. One paragraph is enough. Return the favour and read one other person's draft.
3. A question for the facilitators: if something from Weeks 1 through 5 is still unclear and you want to understand it before Week 6, post it here. Facilitators are monitoring the forum during the break week.

Tip: Reading your peers' capstone ideas is one of the most useful things you can do this break week. Seeing what problems other Kenyan data scientists are choosing to work on will either validate your own direction or reveal an angle you had not considered.

YOUR MENTOR

How to Find Your Mentor

Your mentor's name, organisation, and contact information are on the Moodle "Capstone Mentor Assignments" page. Go there now.

Your First Message

Send within 24 hours of finding your mentor's details:

```
Subject: Ngao Labs Cohort 3 – [Your name] Capstone Introduction
Hello [Mentor name], My name is [your name] and I am in Ngao Labs Cohort 3.
You have been assigned as my Capstone Mentor. I am grateful for your
time. A bit about my background: [one sentence on your professional
or educational background]. My strongest weeks in the programme so
far have been [name one or two modules]. I am currently considering
[brief description of your capstone direction, one to two sentences].
I will send you my full project idea document by the start of Week 6
for your review. My GitHub profile is [link]. I have already created
a repository for the project at [link]. I look forward to working
with you over the next five weeks. Best regards, [Your name] [Your
preferred contact method]
```

Do not send a long first message. The template above is the right length. Your mentor will know what to expect and when to expect the idea document.

Resources

1. Capstone Planning

Week 10 Lesson — Full Capstone Guide (Moodle)

Your most important resource this week. Contains the complete project workflow, problem statement template, GitHub README template, Responsible AI Statement structure, and assessment rubric.

Cookiecutter Data Science — Standard Repository Structure

Read the "Getting started" section to understand the folder structure your capstone repository should follow.

Google ML Problem Framing (Free, 1 Hour)

The clearest single resource for writing a problem statement that is specific, testable, and connected to a real decision.

2. Kenyan and African Datasets to Explore

Zindi Africa — All Competitions

Every challenge dataset is available to download. Filter by "closed" to see the full archive including all six datasets from this programme.

Kenya National Bureau of Statistics Open Data

Census 2019, commodity prices, economic surveys. The most useful source for genuinely Kenyan capstone datasets.

Humanitarian Data Exchange (Kenya)

Food security, population, health, and climate data for Kenya. All datasets are freely downloadable.

3. GitHub Basics

GitHub Docs: Hello World Tutorial

A 10-minute tutorial. Complete it before Week 6 if you have not used GitHub before. Your capstone repository must be public and active from Week 6 onward.

How to Write a Good Git Commit Message

Short. Worth reading before your first commit. Commit messages like "added stuff" or "update" tell a reviewer nothing about what changed. Messages like "Add logistic regression baseline, AUC-ROC 0.71" tell a story of iterative progress.

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